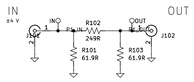
P1 pi pad, −20 dB

cell (0.0)

tutorial On the air, The modem against noise

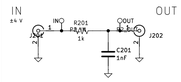


EXPECT: −20 dB flat, 50 ohm both ways
A pi attenuator: 61.9 / 249 / 61.9 ohm is −20.0 dB and 50.0 ohm in a 50 ohm system (the tutorial's $10 SMA pad, on the board).
Into a high-impedance load it is only −14 dB: the pad assumes 50 ohm on both sides.

P2 RC low-pass, 159 kHz

cell (1.0)

tutorial RC and LC circuits, Feedback control of an RC plant, Control at the speed of the cable

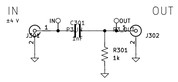


EXPECT: −3 dB at 159 kHz, then −20 dB/decade; OUT into the ADC's high impedance
1 k and 1 nF: f = 1 / (2 pi R C) = 159 kHz. Measure OUT with the ADC on its high-impedance setting: a 50 ohm
load would make a 1 k / 50 divider of it. The lag plant of the tutorial's feedback-control pages: the step response is 1 − exp(−t / 1 us).

P3 RC high-pass, 159 kHz

cell (2.0)

tutorial RC and LC circuits

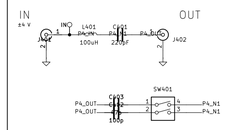


EXPECT: −3 dB at 159 kHz, +20 dB/decade below; OUT into high impedance
The mirror image of P2: the same 1 k and 1 nF the other way round. A step in gives a spike that decays in 1 us.

P4 series RLC, 100uH 220pF

cell (3.0)

tutorial RC and LC circuits, Control at the speed of the cable



the DIP switch and its two capacitors are drawn on the right: way 1 adds 47 pF, way 2 adds 100 pF across C
(each way from the L–C node to the OUT node through its capacitor)

EXPECT: a peak at 1.07 MHz (switches off), 0.98 with 1, 0.89 with 2, 0.83 with both; Q about 6
Series L C between 50 ohm ports: a band-pass at f = 1 / (2 pi sqrt(L C)) whose Q ≈ sqrt(L/C) / (100 + R_coil) is about 6
(the 10 ohm coil would give 60 by itself). The DIP switch adds 47 pF (way 1) and 100 pF (way 2) across the 220 pF to move
the peak: a resonant plant for feedback control, a lockbox's cavity before the real one. No OUT loop (the switch took its corner).

P5 LC low-pass, 9 MHz

cell (0.1)

tutorial Fast captures, Chirps, Zadoff–Chu and LoRa, RC and LC circuits

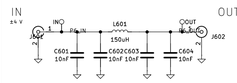


EXPECT: −3 dB at 8.9 MHz, −30 dB at 18 MHz: the anti-alias filter in front of a 25 MS/s ADC
5th-order Butterworth, 50 ohm: 220 pF, 1.5 uH, 680 pF, 1.5 uH, 220 pF (the exact values are 219 pF, 1.43 uH, 707 pF).
Before the ADC it removes the alias (the ghost image above Nyquist): after the DAC it smooths the steps.

P6 LC low-pass, 100 kHz

cell (1.1)

tutorial Trading speed for bits, RC and LC circuits

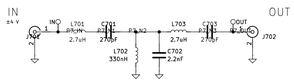


EXPECT: −3 dB at 118 kHz: the reconstruction filter after a one-bit DAC
3rd-order pi, 50 ohm: 2 x 10 nF, 150 uH, 2 x 10 nF, all C0G (JLCPCB stocks no 22 nF C0G in 0805 or 1206).
Turns a one-bit (delta-sigma) DAC stream into a tone: a slow plant for feedback control, with P2.

P7 band-pass, 4.7–7.5 MHz

cell (2.1)

tutorial Chirps, Zadoff–Chu and LoRa, A radio link, On the air

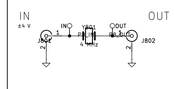


EXPECT: the digital-communications pages' band (a 6.25 MHz carrier): a peak at 6.3 MHz, 3 dB from 4.7 to 7.5 MHz
3rd-order Butterworth band-pass, 6.05 MHz centre, 3.1 MHz wide: series 2.7 uH + 270 pF, shunt 330 nH || 2.2 nF, series 2.7 uH + 270 pF
(the exact values: 2.57 uH, 269 pF, 337 nH, 2.05 nF). The tutorial's filtered chips in copper: it cleans a transmitter's
harmonics and is a radio link's receiver front end.

P8 quartz crystal, 4 MHz

cell (3.1)

tutorial The Q of a quartz crystal, Your crystal against a reference, More ideas for one board



EXPECT: a sharp pass at the series resonance near 4.000 MHz (−3.7 dB) and a notch 6.0 kHz above it, the parallel resonance
The crystal in series between 50 ohm ports: it passes at its series resonance f_s (R_m about 50 ohm: −3.5 dB) and blocks at
the parallel resonance f_p = f_s (1 + C_m / 2 C_0) a few kHz above. A capacitor across OUT (a tee at the ADC) pulls f_p
down toward f_s and leaves f_s alone: the pulling the tutorial measures against a reference. Model: C_m 15 fF, Q 50 000, C_0 4 pF (typical HC-49S).

P9 ceramic res., 455 kHz

cell (0.2)

tutorial RC and LC circuits, The Q of a quartz crystal

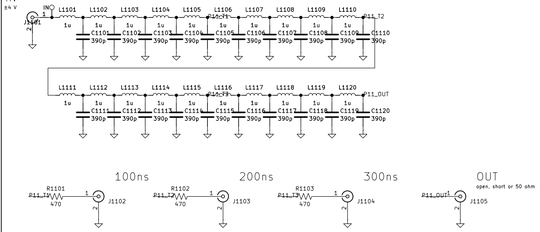


EXPECT: a pass near 455 kHz and a notch a few kHz above: a resonator with a Q of a few hundred, between P4's LC and P8's quartz
A two-terminal 455 kHz ceramic resonator in series (JLCPCB stocks no three-terminal 455 kHz filter). It is the superhet's IF
element, and here a resonance whose Q (hundreds) sits between the LC's and the crystal's. Model: C_m 7 pF, Q 400, C_0 250 pF.

P11 delay line, 400 ns, a tap every 100

cells (1.2), (2.2), (3.2)

tutorial A quarter-wave stub, The channel, Chirps, Zadoff–Chu and LoRa, Control at the speed of the cable

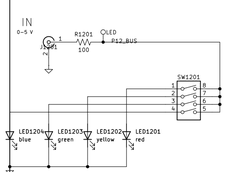


EXPECT: a pulse arrives at OUT 395 ns late, at the taps every 99 ns; flat to about 10 MHz, then it disperses
Twenty sections of 1.0 uH and 390 pF: Z0 = sqrt(L/C) = 51 ohm, 20 ns a section, cut-off 1 / (pi sqrt(LC)) = 16 MHz,
400 ns is 260 ft of RG-58. Leave OUT open and a pulse comes back the same way up: short it and it comes back inverted:
50 ohm on OUT and nothing comes back (the quarter-wave and reflection experiments). The taps are a channel's echo and a
radar target on a cable: near 16 MHz the phase velocity falls and a chirp spreads (below that, a pure-delay plant for control). A tap through 470 ohm: x1 into a
high-impedance scope, /10 into 50 ohm.

P12 LEDs

cell (0.3)

tutorial More ideas for one board

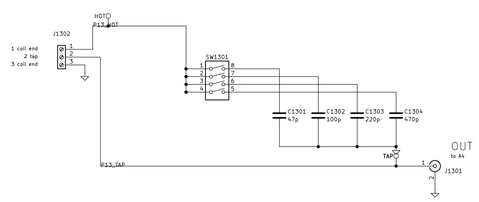


EXPECT: one LED at a time: the voltage where it starts to glow against 1/wavelength is h/e
100 ohm from the jack to the four LEDs: the 4-way switch picks one. Drive from the DAC through the ECP5 board's unipolar
LED output or a bench supply: the knee voltage of each colour against c / lambda gives Planck's constant.
Way 1 red 630 nm (AlGaInP), 2 yellow 596 (AlGaInP), 3 green 525 (InGaN), 4 blue 469 nm (InGaN): the InGaN pair lie
above the line. The LED loop is after the 100 ohm: the LED's own voltage. The colours are not on the silk: light them.

P13 AM loopstick, 0.5–1.7 MHz

cells (1.3), (2.3)

tutorial An AM radio

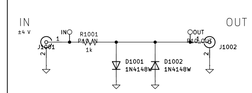


EXPECT: with a 330 uH loopstick: 1.28 MHz on 1, 0.88 on 2, 0.59 on 3, 0.40 MHz on 4, and the sums between (the AM band)
Any ferrite-rod AM antenna coil (about 330 uH: 50 to 100 mm long) screws into the terminal. Its tap (a quarter of the turns
from the ground end) to the jack. The switch's capacitors tune it: f = 1 / (2 pi sqrt(L C)): 1.28 MHz with 47 pF down to
0.40 MHz with 470 pF; combine positions for the steps between. Into AA's LNA and the tutorial's AM radio gateware: a real station.

P10 clipper, 1N4148 ±0.6 V

cell (3.3)

tutorial Harmonics: a diode clipper



EXPECT: a sine of more than 1 V comes out flat-topped at about ±0.6 V; the ADC on high impedance
1 k in series, then a 1N4148W each way to ground: whatever goes in, OUT stays within a diode drop. Harmonics: the
clipped sine's spectrum against the amplitude: the 1 k limits the diode current to a few milliamps from a ±4 V source.

MH1 MH2 MH3 MH4
M3 mounting holes, one per corner

GND GND GND GND
ground loops for the scope's clips, one per channel, in the board's margins

PWR_FLAG

H9901 H9902 H9903 H9904
slots for cable ties

Drawn by scripts/gen_sch.py from boards/amod_p.py: the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
pads. RC and LC filters, resonators, a crystal, a clipper, a tapped delay line, LEDs, an AM loopstick
Board: Jason Gallicchio and Claude | open hardware, CERN-OHL-P-2.0 | github.com/gallicchio/

Sheet: /	
File: amod_p.kicad_sch	
Title: AMOD-P: passive building blocks for the converters	
Size: A1	Date: 2026-10-25
KiCad E.D.A. 10.0.7	**Rev: A**
	Id: 1/1